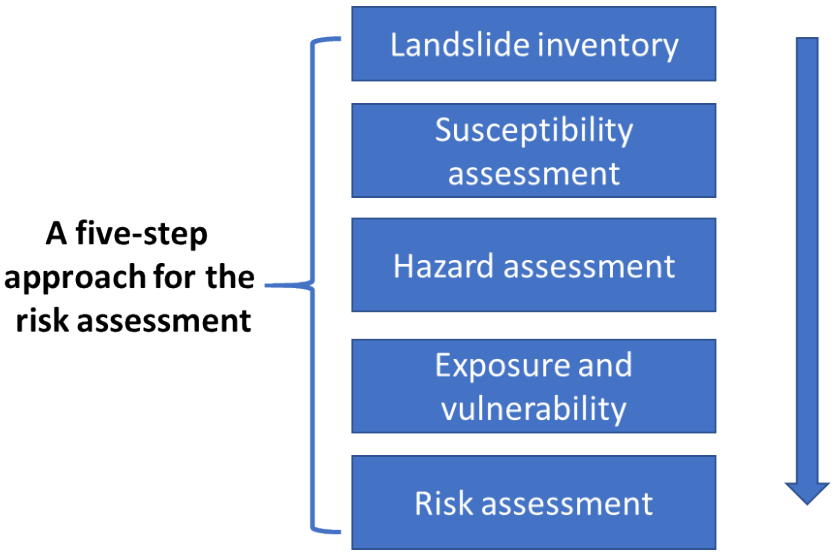


Figure 1 – *Landslide risk assessment methodology*

2.2 Available knowledge and required data

There are several research works on landslide susceptibility, hazard and risk assessment that have already been carried out in Rwanda and in the Lake Kivu region in particular (e.g. Jacobs et al., 2018; Depicker et al., 2020a, 2020b, in review; Dewitte et al., 2021). Our approach is to build the risk assessment as much as possible upon this existing knowledge and the methods that have already proved to be successful.

One important issue related to the risk assessment at the scale of the four sub-catchments is the quality of the ancillary data that are provided. These data are used to inventory the landslides and to derive the environmental factor maps that will be used for the susceptibility analysis. The following ancillary data were used for the for susceptibility, hazard and risk assessment in this project:

- Digital Elevation Model (DEM) – SRTM 30 m resolution (Source: Farr et al. 2007) and Rwanda 10m DEM (Source: GGGI / RWB). Note that the latter does not cover the Kamembe-Gihundwe sub-catchment;
- Soil data (Source: GGGI);
- Land cover and land use 2008 (Source RWB), 2018 (Source: GGGI);
- Mapping of road network and drainage lines (Source: RTDA);
- Fault lineaments (Smets et al., 2016 ; Delvaux et al., 2017);
- Peak ground acceletration – PGA (Delvaux et al., 2017);
- Database of landslides in western Rwanda collected via other research projects from the Royal Museum for Central Africa: locations and timing of the landsides (Depicker et al., 2020a, 2020b ; Dewitte et al., 2021).